

Evaluation Report — 2026-06-24

A side-by-side comparison of four AI coding tools run against an identical set of prompts. Each tool was measured on **cost**, **speed**, and **correctness** across three scenarios: one greenfield build and two brownfield analysis tasks.

Summary

- **Run date:** 2026-06-24
- **Aggregation:** mean over 1 run per tool/scenario
- **Mode:** Automated
- **Tools under test:** Claude Code, Codex CLI, Kiro CLI, Kiro IDE

Headline findings

- **Lowest cost:** Kiro CLI at **\$0.117** total across the three scenarios — roughly 25x cheaper than Codex CLI and 13x cheaper than Claude Code.
- **Fastest:** Codex CLI completed all three scenarios in **186.5s** total (62.2s average), the quickest of the group.
- **Most expensive:** Codex CLI at **\$2.903** total, driven by large input token counts billed at the GPT-5.5 rate.
- **Best cost/speed balance:** Kiro IDE — near Kiro-CLI pricing (\$0.152 total) with the second-fastest aggregate time (385s).

Configuration

Tool	Model	Effort	Mode
Claude Code	us.anthropic.claude-opus-4-8	xHigh	Automated
Codex CLI	openai.gpt-5.5	High	Automated
Kiro CLI	claude-opus-4.8	xHigh	Automated
Kiro IDE	claude-opus-4.8	xHigh	Manual

Methodology

- **Harness:** runs were executed and aggregated by `tools/run_benchmark.py` (run-average mode) and recorded in `run-complete.json`.
- **Procedure:** each tool received the identical prompt for each scenario, run in Automated mode with no manual intervention. Output was written to the tool's designated folder and metrics captured from each tool's own usage reporting.
- **Runs and aggregation:** 1 run per tool/scenario; reported figures are the mean (a single run here, so mean equals the observed value).
- **Metrics captured:**
 - *Cost (USD)* — derived per the pricing and methodology below.
 - *Time (s)* — wall-clock elapsed time from prompt to finished result.

- *Lines* — lines of output produced (where reported by the tool).
- *Tokens* — input/output token counts (where reported by the tool).
- **Held constant:** prompts, scenario order, model effort level (xHigh for Claude/Kiro, High for Codex), and Automated mode across all tools.
- **Not scored:** correctness/quality of output is out of scope for this run and assessed separately.

Pricing

OpenAI GPT 5.5 (on Amazon Bedrock) — used for Codex CLI

Token type	Price per 1M tokens
Input	\$5.50
Cached input	\$0.55
Output	\$33.00

Claude Opus 4.8 (on Amazon Bedrock) — used for Claude Code

Token type	Price per 1M tokens
Input	\$5.00
Output	\$25.00

Kiro credits — used for Kiro CLI and Kiro IDE

Unit	Price
Per credit	\$0.02

Cost methodology

- **Kiro CLI / Kiro IDE:** derived from credits consumed at **\$0.02/credit**.
- **Claude Code:** reported run cost in USD from the `/usage` command (Connected to Amazon Bedrock, Claude Opus 4.8 at \$5.00/1M input and \$25.00/1M output; reported totals also include cache read/write).
- **Codex CLI:** computed from session token totals using the GPT-5.5 on Amazon Bedrock pricing above (input × \$5.50/1M + output × \$33.00/1M).

Evaluated Scenarios

- **GF-1** — Greenfield build: produce a fresh, spec-bound CLI to-do app from scratch.
- **CA-1** — Brownfield analysis: read an existing TypeScript API and explain architecture, pricing, and the order state machine.
- **CA-2** — Brownfield investigation: pinpoint which check rejects a specific request and what response it returns.

Results

Full results matrix

Scenario	Tool	Cost (USD)	Time (s)	Lines	Input Tokens	Output Tokens
GF-1	Claude Code	\$0.3281	87.1	197	2,448	4,079
GF-1	Codex CLI	\$0.7275	62.7	164	106,652	4,270
GF-1	Kiro CLI	\$0.0316	64.8	189	NA	NA
GF-1	Kiro IDE	\$0.0540	138.0	NA	NA	NA
CA-1	Claude Code	\$0.7966	313.2	308	2,577	16,931
CA-1	Codex CLI	\$1.4613	90.1	324	228,014	6,279
CA-1	Kiro CLI	\$0.0518	102.8	281	NA	NA
CA-1	Kiro IDE	\$0.0620	194.0	NA	NA	NA
CA-2	Claude Code	\$0.4145	61.3	111	2,446	4,745
CA-2	Codex CLI	\$0.7140	33.7	96	115,527	2,383
CA-2	Kiro CLI	\$0.0336	296.0	88	NA	NA
CA-2	Kiro IDE	\$0.0360	53.0	NA	NA	NA

Token counts are not reported for Kiro CLI/IDE (credit-based billing). Kiro IDE does not report a line count in this run.

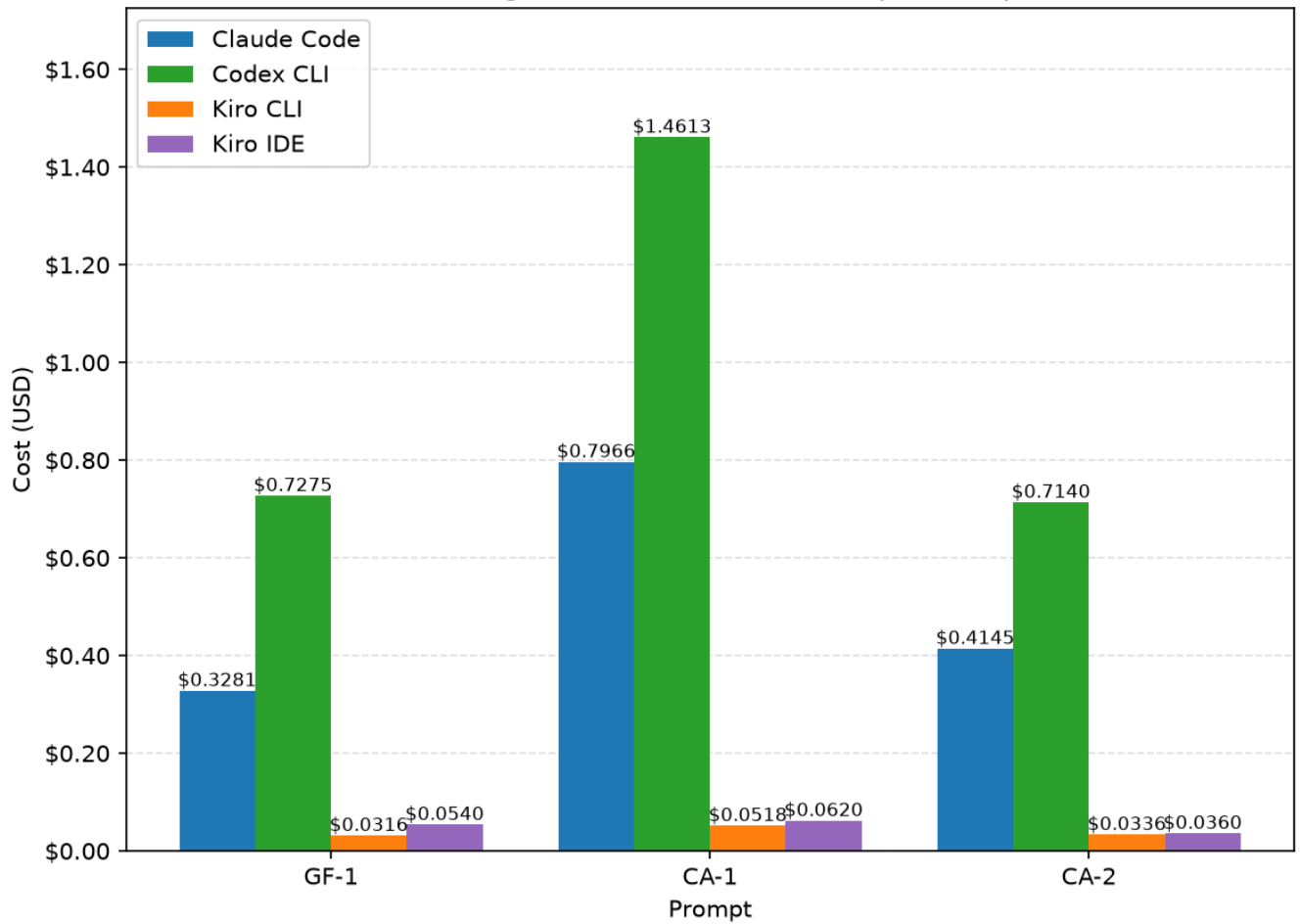
Aggregate by tool

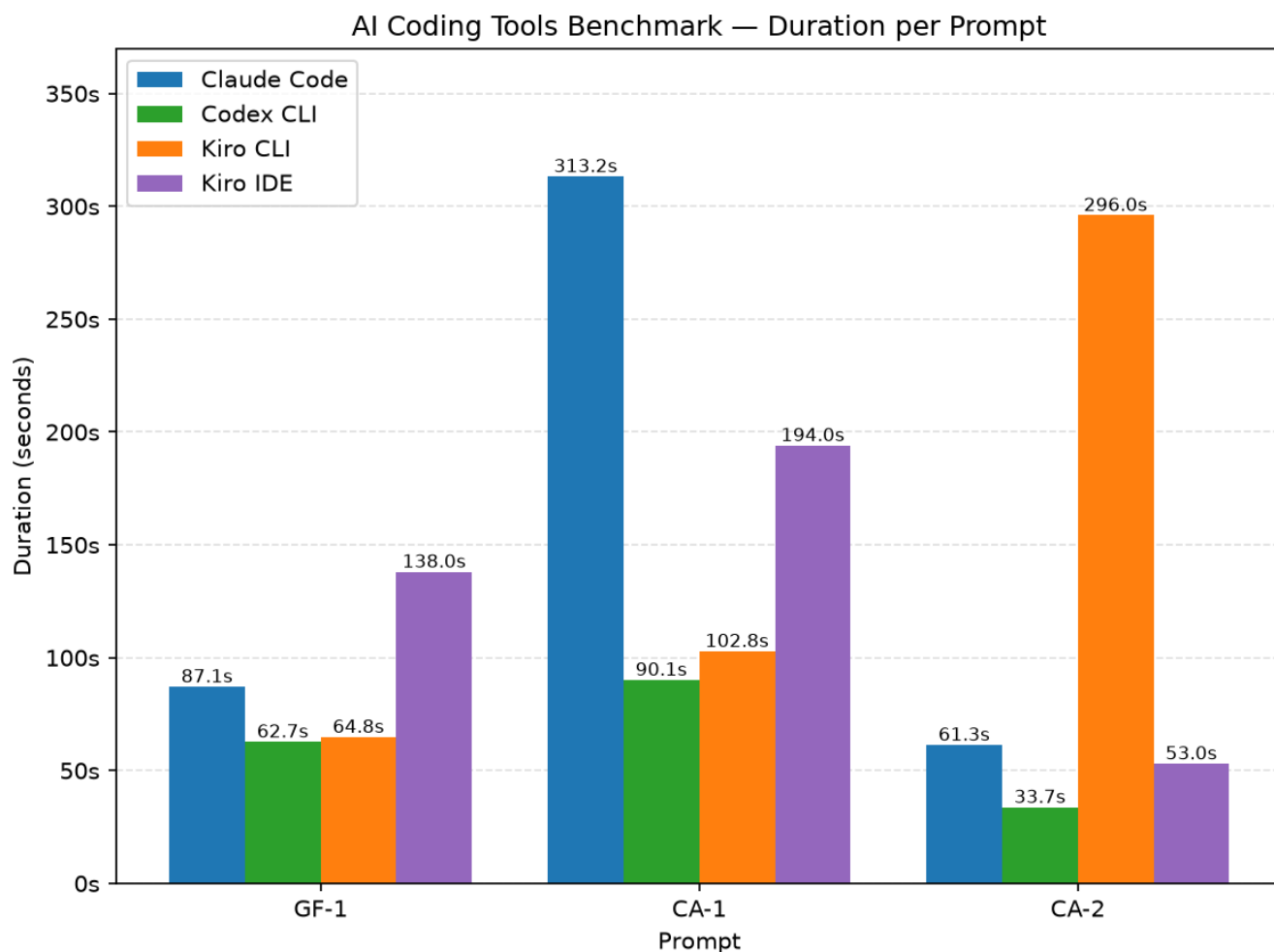
Totals and means across all three scenarios:

Tool	Total Cost (USD)	Avg Cost (USD)	Total Time (s)	Avg Time (s)	Total Lines
Kiro CLI	\$0.1170	\$0.0390	463.6	154.5	558
Kiro IDE	\$0.1520	\$0.0507	385.0	128.3	NA
Claude Code	\$1.5392	\$0.5131	461.6	153.9	616
Codex CLI	\$2.9028	\$0.9676	186.5	62.2	584

Visualizations

AI Coding Tools Benchmark — Cost per Prompt





Per-Scenario Breakdown

GF-1 — Greenfield build

All four tools produced comparable output volume (164–197 lines). On cost, Kiro CLI (\$0.0316) and Kiro IDE (\$0.0540) were an order of magnitude cheaper than Claude Code (\$0.3281) and Codex CLI (\$0.7275). Codex CLI was the fastest to complete after Kiro CLI, but its large input token footprint (106,652 tokens) made it the most expensive on this scenario.

CA-1 — Brownfield analysis

This was the most demanding scenario and the widest cost spread. Codex CLI cost \$1.4613 — the single most expensive result in the run — driven by 228,014 input tokens. Claude Code followed at \$0.7966 with the highest output token count (16,931). Codex CLI was fastest (90.1s) despite the cost, while Kiro CLI delivered the analysis for \$0.0518, roughly 28x cheaper than Codex CLI.

CA-2 — Brownfield investigation

The lightest scenario by output. Codex CLI was again the fastest (33.7s) and the most expensive among the paid-token tools (\$0.7140). Kiro CLI was the slowest here (296.0s) but the cheapest (\$0.0336). Kiro IDE offered a strong balance: \$0.0360 in 53.0s.

Observations

- **Credit-based tools dominate on cost.** Kiro CLI and Kiro IDE were consistently 10–30x cheaper per scenario than the token-billed tools, because Codex CLI in particular sends very large input contexts (over 100K tokens per scenario) billed at \$5.50/1M.
- **Codex CLI trades cost for speed.** It was the fastest tool overall (62.2s average) but also the most expensive (~\$0.97 average per scenario).
- **Claude Code sits in the middle on speed** but its cost scales with output volume — CA-1's 16,931 output tokens pushed it to \$0.7966.
- **Kiro CLI's time is uneven.** It was competitive on GF-1 and CA-1 but slow on CA-2 (296.0s), which inflated its aggregate time above Kiro IDE despite a lower total cost.

Caveats

- Results reflect a **single run** per tool/scenario; figures are means but not yet variance-tested. Treat individual numbers as indicative rather than definitive.
- Kiro IDE line counts and Kiro token counts are unavailable due to how each tool reports usage, limiting like-for-like comparison on those dimensions.
- This report measures cost, speed, and output volume only. **Correctness of the output is not scored here** and should be assessed separately before drawing quality conclusions.

References

- Source repository: <https://github.com/aws-samples/sample-ai-coding-tools-benchmark>